CH 5 - Dimension and Coordiantes

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Bases and Dimension

Exmaple:

Let $\mathbb{S} = \text{Span}\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}-1\\3\\1\end{bmatrix}, \begin{bmatrix}2\\-1\\2\end{bmatrix}\right\}$, find a basis for $\mathbb{S}$

Let $A = \begin{bmatrix}1 & -1 & 2\\2 & 3 & -1\\3 & 1 & 2\end{bmatrix}$. Then $\mathbb{S} = \text{Col}(A)$

$\begin{bmatrix}1 & -1 & 2\\2 & 3 & -1\\3 & 1 & 2\end{bmatrix} \sim R_2 - 2R_1, R_3 - 3R_1 \begin{bmatrix}1 & -1 & 2\\0 & 5 & -5\\0 & 4 & -4\end{bmatrix} \sim \ldots \begin{bmatrix}1 & 0 & 1\\0 & 1 & -1\\0 & 0 & 0\end{bmatrix}$

$\text{Col}(A) = \text{Span}\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}-1\\3\\1\end{bmatrix}\right\}$ is a basis for $\mathbb{S}$

> **Info — Existence of Bases**
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> Every subspace $\mathbb{S}$ of $\mathbb{R}^n$ has a basis.

> **Info — Dimensions**
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> 1. Let $\mathbb{S}$ be a subspace of $\mathbb{R}^n$. Let $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be a basis for $\mathbb{S}$ and let $C = \{\vec{w}_1, \ldots, \vec{w}_w\}$ be a set in $\mathbb{S}$. If $m > k$, then C is linearly dependent
> 2. If $B = \{\vec{v_1}, \ldots, \vec{v_k}\}$ and $C = \{\overrightarrow{w_1}, \ldots, \overrightarrow{w_m}\}$ are bases for $\mathbb{S}$, then $k = m$.
> 3. Let $\mathbb{S}$ be a subspace of $\mathbb{R}^n$. If $B = \{\vec{v_1}, \ldots, \vec{v}_k\}$ is a basis for $\mathbb{S}$, then we say:
>
> $$\dim \mathbb{S} = k$$
>
>   If $\mathbb{S} = \left\{\vec{0}\right\}$ is a trivial subspace, then $\dim \mathbb{S} = 0$.
> 4. If $\dim \mathbb{S} = k$
>   1. A set of more than k vectors in $\mathbb{S}$ must be linearly dependent.
>   2. A set of more than k vectors in $\mathbb{S}$ cannot span $\mathbb{S}$.
>   3. A set of k vectors in $\mathbb{S}$ is linearly independent if and only if it spans $\mathbb{S}$
> 5. If $\mathbb{S}$ is a k dimensional subspace of $\mathbb{R}^n$ and $\{\vec{v}_1, \ldots, \vec{v}_l\}$ is a linearly independent set in $\mathbb{S}$ with $l < k$, then there exist vectors $\vec{w}_{l+1}, \ldots, \vec{w}_k \in \mathbb{S}$ s.t. $\{\vec{v}_1, \ldots, \vec{v}_l, \vec{w}_{l+1}, \ldots, \vec{w}_k\}$ is a basis for $\mathbb{S}$.
> 6. Let $\mathbb{S}_1, \mathbb{S}_2$ be subspace of $\mathbb{R}^n$ s.t. $\mathbb{S}_1 \subseteq \mathbb{S}_2$. Then $\dim \mathbb{S}_1 \leq \dim \mathbb{S}_2$. Moreover $\mathbb{S}_1 = \mathbb{S}_2$ if and only if $\dim \mathbb{S}_1 = \dim \mathbb{S}_2$

Proof

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Since B is a basis for $\mathbb{S}$ and $\vec{w}_i \in \mathbb{S}\ \forall i = 1, ..., m,\ \exists$ coefficients a_{ij} s.t.

$$\begin{aligned}\vec{w}_1 &= a_{11}\vec{v}_1 + a_{12}\vec{v}_2 + ... + a_{1k}\vec{v}_k \\ \vec{w}_2 &= a_{21}\vec{v}_1 + a_{22}\vec{v}_2 + ... + a_{2k}\vec{v}_k \\ ... &= ... \\ \vec{w}_m &= a_{m1}\vec{v}_1 + a_{m2}\vec{v}_2 + ... + a_{mk}\vec{v}_k\end{aligned}$$

Consider
$\vec{0} = c_1\vec{w}_1 + c_2\vec{w}_2 + ... + c_m\vec{v}_m$

$$\begin{aligned}\vec{0} &= c_1(a_{11}\vec{v}_1 + a_{12}\vec{v}_2 + ... + a_{1k}\vec{v}_k) + c_2(a_{21}\vec{v}_1 + a_{22}\vec{v}_2 + ... + a_{2k}\vec{v}_k) + ... + c_m(a_{m1}\vec{v}_1 + a_{m2}\vec{v}_2 + ... + a_{mk}\vec{v}_k) \\ &= (c_1a_{11} + c_2a_{21} + ... + c_ma_{m1})\vec{v}_1 + (c_1a_{12} + c_2a_{22} + ... + c_ma_{m2})\vec{v}_2 + ... + (c_1a_{1k} + c_2a_{2k} + ... + c_ma_{mk})\vec{v}_k\end{aligned}$$

Since B is a basis,

$$\begin{aligned}(c_1a_{11} + c_2a_{21} + ... + c_ma_{m1}) &= 0 \\ (c_1a_{12} + c_2a_{22} + ... + c_ma_{m2}) &= 0 \\ ... &= ... \\ (c_1a_{1k} + c_2a_{2k} + ... + c_ma_{mk}) &= 0\end{aligned}$$

This is a homogenous linear system with coefficient matrix $A = \begin{bmatrix} a_{11} & a_{21} & ... & a_{m_1} \\ ... & ... & ... & ... \\ a_{1k} & a(2k) & ... & a_{mk} \end{bmatrix}$

A is a $k \times n$ matrix. Since $m > k$, there are more columns than rows.

There are at least one free variable $\implies$ infinitely many solutions

$\implies c = \{\vec{w}_1, ..., \vec{w}_m\}$ is linearly dependent.

□

Info — Coordinates

1. If $B = \{\vec{v}_1, ..., \vec{v}_k\}$ is a basis for a subspace $\mathbb{S}$ of $\mathbb{R}^n$, then every $\vec{v} \in \mathbb{S}$ can be written as a unique linear combination of the vectors in B
2. Let $B = \{\vec{v}_1, ..., \vec{v}_k\}$ be a basis for a subspace $\mathbb{S}$ of $\mathbb{R}^n$. If $\vec{v} = b_1\vec{v}_1 + ... + b_k\vec{v}_k$, then $b_1, ..., b_k$ are called the **B-coordinates** of $\vec{v}$, and we define the **B-coordinate vector** by
$$[\vec{v}]_B = \begin{bmatrix} b_1 \\ ... \\ b_k \end{bmatrix}$$
3. If $\mathbb{S}$ is a subspace of $\mathbb{R}^n$ with basis $B = \vec{v}_1, ..., \vec{v}_k$, then for any $\vec{v}, \vec{w} \in \mathbb{S}$, $s, t \in \mathbb{R}$
$$[s\vec{v} + t\vec{w}]_B = s[\vec{v}]_B + t[\vec{w}]_B$$

Note: the order of basis matter(i.e. $\{\vec{e}_1, \vec{e}_2\} \neq \{\vec{e}_2, \vec{e}_1\}$)

Concept:

For an arbitrary $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ ... \\ x_n \end{bmatrix} \in \mathbb{R}^n$, we can write $\vec{x} = x_1\vec{e}_1 + ... + .x_n\vec{e}_n$

Which leads to Theorem 2.

Info — Change of Coordinates Matrix

1. Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$ and C both be bases for a subspace $\mathbb{S}$. The change of coordinates matrix from B —coordinates to C-coordinates is defined as

   $$_CP_B = [[\vec{v}_1]c...[\vec{v}_n]c]$$

   and for any $x \in \mathbb{S}$, we have

   $$[\vec{x}]_c = {}_CP_B[\vec{x}]_B$$

2. If B and C are bases for a k-dimensional subspace $\mathbb{S}$, then the change of coodinates matrices $_CP_B$, $_BP_C$ satisfy

   $$_CP_B{}_BP_C = I = {}_BP_C{}_CP_B$$